H.K.E SOCIETY'S

POOJYA DODDAPA APPA COLLEGE OF ENGINEERING

KALABURGI 585-102

(An Autonomous institute affiliated to VTU Belagavi, and approved by AICTE)

A

INTERNSHIP REPORT

ON

"BOOK MY TICKET"

Submitted to the

POOJYA DODDAPPA APPA COLLEGE OF ENGINEERING KALABURAGI,

An Autonomous Institution, Affiliated to VTU Belagavi and Approved by AICTE in the Partial Fulfilment of the requirements for the Award of Degree of

BACHELOR OF ENGINEERING IN

COMPUTER SCIENCE AND DESIGN

Submitted by

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Under the guidance of

Dr ANITA HARSOOR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

P.D.A COLLEGE OF ENGINEERING (AUTONOMOUS INSTITUTION),

KALABURAGI 2025-2026

HYDERABAD KARNATAKA EDUCATION SOCIETY'S

POOOJYA DODDAPPA APPA COLLEGE OF ENGINEERING,

KALABURGI 585-102

(An Autonomous institute affiliated to VTU Belagavi, and approved by AICTE)

ESTD : 1958

CERTIFICATE

This is to certify that the Internship on " BOOK MY TICKET" carried out by MADIHA AZAM (3PD22CG048 in partial fulfillment for the award of Bachelor of Engineering in Computer Science & Engineering of Poojya Doddappa Appa College of Engineering, Kalaburagi, an Autonomous Institution, affiliated to Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. The internship report has been approved as it satisfies the academic requirements in respect of work prescribed.

Dr. ANITA HARSOOR	**Dr. SHRIDEVI SOMA**	**Dr. S.R. PATIL**
Guide	*Head of the department*	*Principal*

Examiners :

1.

2.

DECLARATION

I **MADIHA AZAM , USN NO : 3PD22CG048** student of Poojya Doddappa Appa College of Engineering, Kalaburagi here by declare that the dissertation entitled **"BOOK MY TICKET"** embodies the report of my internship carried out independently by me during **8th sem of Bachelor of Engineering Department of Computer Science and Engineering Kalaburagi** and this work has been submitted for the fulfilment of the requirement for the award of BE degree.

I Have Not submitted the matter embodies to any other university or institution for the award of any other degree.

DATE: **MADIHA AZAM**

PLACE: **(3PD22CG048)**

ACKNOWLEDGEMENT

We express our deep sense of gratitude and indebtness to our esteemed institute "PDA COLLEGE OF ENGINEERING", KALABURAGI which has provided us an opportunity to fulfil the most cherished desire to reach our goal.

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MADIHA AZAM 3PD22CG048

ABSTRACT

This internship report presents the work carried out by **MADIHA AZAM (3PD22CG048)** during a 15-week Full Stack Web Development (JAVA) internship at Jspiders, Bengaluru, in partnership with Visvesvaraya Technological University (VTU), Belgaum, from January 22 2026 to May 22th 2026. This project report presents the design and development of the “BOOK MY TICKET” system, a web-based online ticket booking application developed using modern Java technologies and frameworks. The project aims to provide an efficient digital platform that enables users to book tickets for movies, buses, trains, flights, or events seamlessly through a centralized system. The application was developed using technologies such as Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript. The system follows the MVC architecture and implements RESTful services for smooth communication between frontend and backend components. The platform provides separate modules for Admin and User operations, ensuring secure and role-based access control. The BOOK MY TICKET system includes functionalities such as user registration and login, ticket search, seat selection, online booking, payment handling, booking history, and cancellation management. Users can search available tickets, select seats, book tickets online, and manage their booking details efficiently. Administrators can manage schedules, ticket availability, customer.

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CHAPTER 1

INTRODUCTION

CHAPTER 1 : INTRODUCTION

The rapid growth of digital technology and online booking platforms has significantly transformed the way transportation and travel services are managed. Traditional ticket booking methods that rely on manual reservations, paper tickets, and offline customer handling have become inefficient due to increasing passenger demands, large volumes of booking requests, and the need for faster and more accurate services. In today’s fast-moving digital environment, organizations require automated and centralized systems that can simplify ticket booking operations while improving efficiency, reliability, and customer satisfaction.

In response to these challenges, the “BOOK MY TICKET” system was developed as a full-stack web application using modern web development technologies and frameworks. The system aims to provide an efficient, secure, and user-friendly platform for managing various ticket booking operations such as user registration, ticket reservation, seat availability checking, booking management, payment handling, and cancellation services through a centralized digital solution.

The internship was conducted through a structured Full Stack Web Development training programme at JSpiders, Bengaluru, in collaboration with Visvesvaraya Technological University (VTU), Belagavi. The training combined theoretical learning with practical implementation, allowing students to gain real-world exposure to frontend development, backend programming, database management, API integration, and responsive web application development. This practical learning approach helped bridge the gap between academic concepts and industry requirements.

The BOOK MY TICKET system was developed using technologies such as Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application follows the MVC (Model-View-Controller) architecture and uses RESTful APIs for smooth communication between frontend and backend components. The system provides separate modules for administrators and users, ensuring secure role-based access and efficient management of booking operations.

The primary objective of this project was to automate and simplify ticket booking activities while reducing manual effort and improving operational efficiency. The system allows administrators to manage routes, schedules, ticket availability, bookings, and customer details efficiently, while users can search available tickets, select destinations, book seats, make payments, and manage their booking history through an interactive interface. Automated booking confirmation, real-time seat availability updates, secure authentication, and centralized data management help improve accuracy and reliability in ticket management operations.

During the development of the project, emphasis was placed on creating responsive and dynamic user interfaces that can function smoothly across desktops, tablets, and mobile devices. The frontend technologies were used to build attractive and user-friendly pages, while backend technologies handled business logic, server-side operations, authentication, and database communication. The integration of frontend, backend, and database technologies provided a complete understanding of modern web application development workflows.

Another important aspect of the internship was the exposure to industry-oriented software development practices. The project encouraged the use of clean coding standards, modular architecture, debugging techniques, API testing, database optimization, and version control concepts. These practices are essential in professional software development environments where scalability, maintainability, and performance are highly important.

The internship also helped in improving analytical thinking, problem-solving ability, teamwork, and independent learning skills. Completing assignments, handling project modules, debugging errors, and integrating multiple technologies within deadlines contributed to the development of a professional and disciplined approach toward software engineering.

Overall, the “BOOK MY TICKET” internship project provided valuable practical experience in full-stack web development and software engineering concepts. The project successfully demonstrated how modern technologies can be used to build scalable, efficient, and user-friendly web applications for online ticket booking and management. The knowledge and skills gained through this internship will be beneficial for future academic projects, technical interviews, and professional opportunities in the field of software development and web technologies.

1.1 Problem Statement

In today's fast-paced transportation environment, passengers and travel agencies face significant challenges in managing ticket booking and reservation operations efficiently. Many organizations still rely on manual methods such as handwritten records, offline reservation systems, and traditional booking counters to maintain passenger details, booking records, schedules, and payment transactions. These outdated approaches often lead to errors, data redundancy, loss of important information, and inefficient ticket management processes.

Managing ticket reservations manually makes it difficult for administrators to track seat availability, monitor bookings, and update travel schedules in real time. As the number of passengers and booking requests increases, maintaining accurate records becomes more complex and time-consuming. Issues such as duplicate bookings, incorrect ticket generation, delayed confirmations, payment errors, and cancellation management negatively affect operational efficiency and customer satisfaction.

Traditional ticket booking systems also lack centralized access to booking information and do not provide advanced features such as automated reservation processing, online payments, booking history management, cancellation tracking, and secure user authentication. Administrators often face difficulties in generating accurate reports related to bookings, revenue, seat occupancy, and customer transactions, which impacts effective decision-making and business growth.

Furthermore, many existing systems are not user-friendly, responsive, or accessible across different devices, limiting their usability in modern digital environments. There is a growing need for a centralized, efficient, and easy-to-use Ticket Booking Management System that can automate reservation operations, manage schedules effectively, generate tickets instantly, and provide secure access to authorized users.

The proposed "BOOK MY TICKET" system aims to address these challenges by providing a digital platform that streamlines ticket booking operations through passenger management, schedule handling, online reservation processing, billing automation, cancellation services, and real-time data management. The system will enhance operational efficiency, reduce manual effort, improve booking accuracy, and support better decision-making for administrators and service providers.

1.2 Proposed System

The proposed "BOOK MY TICKET" system is a web-based application developed to automate and manage various ticket booking and reservation operations efficiently. The project was developed at JSpiders as part of practical learning and implementation of full-stack web development technologies. The system provides a centralized platform for managing ticket reservations, passenger details, schedules, bookings, payments, and transaction records in a secure and organized manner. It is designed to reduce manual work, improve operational efficiency, and enhance the overall booking experience for both administrators and users.

The system enables administrators to add, update, delete, and manage travel schedules, routes, seat availability, pricing details, and booking records. The BOOK MY TICKET system also includes customer management and reservation processing functionalities. Users can search available tickets, select destinations, check seat availability, book tickets, and manage their reservations easily through a user-friendly interface. The system securely stores customer and booking information, making transaction management more efficient and reliable

An integrated billing and payment module allows automatic ticket generation and transaction recording. The system can generate reports related to bookings, payments, seat occupancy, schedules, and customer activities, helping administrators monitor system performance and make informed decisions. To ensure security and controlled access, the proposed system provides secure user authentication and role-based authorization for administrators and users. The application is designed with a responsive interface, enabling smooth access across desktops, laptops, tablets, and smartphones.

Overall, the proposed BOOK MY TICKET system offers a complete digital solution for managing ticket reservation operations efficiently by improving accuracy, reducing manual effort, saving time, and enhancing customer satisfaction.

The system was developed using modern web technologies including Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application follows the MVC (Model-View-Controller) architecture to maintain a clear separation between the user interface, business logic, and database operations. RESTful APIs are implemented to ensure smooth communication between frontend and backend components, allowing efficient data exchange and improved application performance.

The proposed system provides role-based access control with separate functionalities for administrators and users. Each user role is provided with authorized access to specific modules and operations within the application. This helps maintain security, data privacy, and efficient management of booking operations.

The Administrator module is responsible for managing the entire system. Administrators can add, update, and delete schedules, manage ticket availability, monitor customer activities, track bookings, view transaction reports,

1.3 Motivation

The motivation behind developing the “BOOK MY TICKET” system arises from the growing need for an efficient, fast, and reliable digital solution for managing ticket booking operations in today’s transportation and travel industry. Traditional booking methods that depend on manual counters, paper-based records, and offline communication are increasingly inefficient in handling large volumes of passengers and real-time booking demands.With the rapid growth of online services, users expect quick ticket availability checks, instant booking confirmation, and secure payment processing. However, many existing systems still suffer from issues such as booking delays, seat unavailability errors, lack of real-time updates, and poor user experience. These limitations highlight the need for a centralized system that can automate and simplify the entire ticket reservation process.

Another key motivation is to reduce the workload of administrators and service providers by minimizing manual intervention. Automating tasks such as schedule management, seat allocation, booking confirmation, and report generation helps improve accuracy and efficiency while reducing human errors.From a learning perspective, the project also serves as an opportunity to apply full-stack development skills in a real-world scenario. It helps in understanding how frontend, backend, database, and API services work together in an integrated system using technologies like Java, Spring Boot, Hibernate/JPA, and MySQL.Additionally, improving customer satisfaction is a major driving factor. A user-friendly, responsive, and secure platform ensures that customers can easily book tickets anytime and anywhere, leading to a smoother and more reliable experience.Overall, the motivation of the “BOOK MY TICKET” system is to bridge the gap between traditional booking methods and modern digital solutions by creating a scalable, efficient, and user-centric platform for ticket reservation and management.

CHAPTER 2

OVERVIEW OF INTERNSHIP ORGANIZATION

, The internship was conducted at JSpiders, a well-known training and development institute operated under Test Yantra Software Solutions Pvt Ltd. JSpiders is recognized for providing industry-oriented technical training programmes that focus on bridging the gap between academic learning and practical software development skills. The organization offers professional training in various domains such as Full Stack Web Development, Java Technologies, Python Programming, Testing, Data Science, and Frontend Frameworks, helping engineering students and graduates prepare for careers in the IT industry.

JSpiders provides a structured learning environment where students can gain practical exposure to modern software development technologies through classroom sessions, hands-on coding practice, assignments, mini-projects, and real-time application development. The institute emphasizes practical implementation alongside theoretical understanding, enabling students to develop technical skills that are directly relevant to industry requirements.

The primary objective of JSpiders is to transform students into industry-ready professionals by providing high-quality technical education and practical training. Many students learn programming concepts and computer science subjects during their academic studies, but they may not receive sufficient exposure to real-world development tools, coding standards, project workflows, and software engineering practices. JSpiders addresses this gap by offering practical training programmes designed according to current industry standards and technologies.

The training programmes conducted by JSpiders follow an industry-oriented and outcome-based approach. Each module is carefully structured to ensure that students first understand the core concepts and then apply them through practical implementation. This method allows learners to strengthen both conceptual knowledge and programming skills simultaneously. Students are encouraged to solve coding problems, complete assignments, develop projects, and participate in technical discussions to improve analytical thinking and problem-solving abilities.

JSpiders provides training in important technologies that are widely used in modern software development. During the internship programme, emphasis was placed on Full Stack Web Development using technologies such as Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript. These

Organizational Structure of the Program

The internship program was organized into three distinct but complementary modules, delivered in sequence. Each module comprised multiple chapters covering specific technologies or conceptual areas, with each chapter accompanied by practical exercises, mini-projects, or assignments to reinforce the learned concepts. The program concluded with a final assessment and project that integrated knowledge from all three modules.

The program was overseen by a faculty guide assigned by P.D.A College of Engineering, who monitored progress, reviewed submissions, and provided academic guidance throughout the internship period. Regular interaction between students and the faculty guide ensured that the internship activities were aligned with the academic objectives and institutional requirements.

The programme mainly consisted of three important domains: Full Stack Web Development, Python Essentials and Libraries for Data Science, and React JS. These domains were selected because they cover major areas of modern software development. Together, they provided students with knowledge of web technologies, programming fundamentals, data handling, and modern front-end development.

Each module was further divided into chapters or units that covered individual topics in detail. For example, the web development section included topics related to HTML, CSS, JavaScript, Bootstrap, PHP, MySQL, and deployment. Similarly, the Python module included programming basics, data structures, libraries, and visualisation concepts, while the React JS module focused on components, props, state, events, and dynamic rendering.

The sequence of the modules was planned in a way that supported progressive learning. Students first learned foundational technologies and basic programming concepts, then moved toward more advanced tools and frameworks. This order helped in building a strong base before introducing complex development practices.

Practical exercises were included after each topic to help students apply what they had learned. These exercises allowed students to test their understanding, identify mistakes, and improve their coding skills. The hands-on tasks played an important role in converting theoretical knowledge into practical ability.

Assignments and mini-projects formed an essential part of the programme structure. They were designed to encourage independent problem-solving and real-world application of concepts. Through these tasks, students learned how to plan, write, debug, and improve their code in a structured manner.

The programme also included regular monitoring and evaluation. Student progress was reviewed through submitted assignments, completed tasks, and overall participation. This helped ensure that learners were following the schedule and gaining the required understanding from each module.

2.1 Duration and Mode

The internship was conducted during the academic year 2025–2026 at JSpiders as part of the Full Stack Web Development training programme. The internship followed a structured and industry-oriented training schedule designed to provide students with both theoretical understanding and practical exposure to modern web development technologies. The programme focused on helping students strengthen their programming skills, software development knowledge, and practical implementation abilities through regular training sessions, assignments, coding exercises, and project development activities.

The internship programme was conducted in an offline classroom-based learning environment at JSpiders, Bengaluru. Students attended regular technical sessions and practical labs under the guidance of experienced trainers and mentors. The training schedule was organized systematically to ensure proper coverage of frontend technologies, backend development, database management, authentication systems, RESTful APIs, and project implementation concepts. The structured learning process helped students maintain consistency and discipline throughout the internship duration..

Traning module overview		
Module	**Topics Covered**	**Mode**
Frontend Development	HTML, CSS, Bootstrap, JavaScript	Offline
Backend Development	Java, Spring Boot, Hibernate/JPA, REST APIs	Offline
Database Management	MySQL, Database Connectivity, SQL Queries	Offline
Full Stack Integration	Authentication, CRUD Operations, MVC Architecture	Offline
Project Development	BOOK MY TICKET Implementation	Offline

CHAPTER 3

PLAN OF INTERNSHIP TRAINING

3.1 Learning Outcomes

The development of the "E-SHOP Management System" during the internship at JSpiders provided valuable practical exposure to full-stack web application development and modern software engineering practices. The internship helped in strengthening both theoretical understanding and practical implementation skills by allowing real-time development of a complete web-based application. Through continuous learning, coding practice, project implementation, testing, and debugging activities, several technical, analytical, and professional skills were acquired throughout the internship period.

One of the major learning outcomes of the internship was gaining practical experience in full-stack web development using modern technologies such as Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript. The project helped in understanding how frontend technologies are used to design responsive and user-friendly interfaces, while backend technologies manage server-side logic, API communication, authentication, and database operations. Developing the E-SHOP Management System provided a clear understanding of how different layers of a web application interact with each other to deliver complete business functionality.

The internship also improved knowledge of frontend development and responsive web design. Through the implementation of various user interfaces, forms, navigation systems, dashboards, and product management pages, valuable experience was gained in designing attractive and interactive web pages. The project emphasized the importance of responsive design practices to ensure compatibility across desktops, laptops, tablets, and mobile devices. Skills related to layout design, component organization, navigation flow, and user accessibility were significantly improved during the development process

The internship significantly improved understanding of secure authentication and authorization mechanisms. User registration, login systems, password encryption, role-based access control, and session management were implemented to ensure application security. These activities helped in learning how secure authentication systems protect sensitive business and customer information while allowing controlled access to authorized users.

3.3 Attendance

Regular attendance was maintained throughout the internship period to ensure active participation in all training sessions, coding activities, technical discussions, and project implementation tasks conducted at JSpiders.

Consistent attendance played an important role in successfully understanding technical concepts, completing assigned tasks, and gaining practical exposure to full-stack web development technologies and software engineering practices.

3.4 Feedback

The development and implementation of the “E-SHOP Management System” received positive feedback based on its functionality, usability, responsiveness, performance, and practical business application. The project was appreciated for successfully providing a centralized and automated platform that simplifies shop management activities such as inventory management, billing generation, customer handling, order processing, and sales tracking through an organized and user-friendly interface.

One of the most appreciated aspects of the project was its responsive and easy-to-use interface. Users found the application simple to navigate because of its clean design, structured layout, and organized management features. The responsive frontend design ensured smooth accessibility across desktops, laptops, tablets, and mobile devices, which improved user experience and convenience.

The product management and inventory tracking functionalities were also highly appreciated. The ability to add, update, search, categorize, and manage products efficiently helped simplify inventory operations and reduce manual workload. Real-time inventory updates improved stock management accuracy and prevented issues such as duplicate entries and stock shortages.

The secure authentication and authorization system implemented in the application received positive recognition for ensuring controlled access and data protection. Features such as secure login, role-based access control, and protected routes helped improve application security and allowed administrators and users to manage operations safely.

CHAPTER 4

TRAINING PROGRAM

4.1 Methodology

Requirement Analysis

- The first phase of development involved identifying the objectives and requirements of the system.
- Functional and non-functional requirements were analyzed before starting implementation.
- The major requirements identified were:
  - Efficient product and inventory management
  - Customer registration and secure authentication
  - Product search and category filtering
  - Automated billing and order management
  - Sales and transaction record maintenance
  - Responsive and user-friendly interface
- Requirement analysis helped in understanding the overall workflow and business needs of the application.

System Design

- After requirement analysis, the system architecture and workflow were designed.
- The application followed a three-tier architecture consisting of:
  - Frontend layer
  - Backend layer
  - Database layer
- The frontend was designed for interactive user interfaces.
- The backend handled APIs, business logic, validation, and server-side operations.
- The database was designed to store products, customers, orders, inventory, and billing information.
- API flow and database schemas were planned carefully for smooth communication between modules.

Frontend Development

- The frontend was developed using HTML, CSS, Bootstrap, and JavaScript.
- Major frontend development activities included:
  - Designing reusable UI components
  - Creating responsive layouts
  - Implementing routing and navigation
  - Developing shopping cart and billing pages
  - Designing product display and search modules
  - Integrating frontend with backend APIs
- Responsive design techniques ensured compatibility across desktops, tablets, and mobile devices.

Backend Development

- The backend was developed using Java, Spring Boot, and Hibernate/JPA.
- RESTful APIs were created for frontend-backend communication.
- Backend functionalities included:
  - User authentication and authorization
  - Product and inventory management

  - Customer and order handling
  - Billing and transaction processing
  - Validation and exception handling
- The backend acted as an intermediary between frontend and database layers.

Database Integration
- MySQL was used for storing and managing application data.
- Separate tables were created for:
  - Users
  - Products
  - Inventory
  - Orders
  - Billing and transactions
- Hibernate/JPA was used for object-relational mapping.
- Database relationships were designed carefully to maintain:
  - Data consistency
  - Efficient retrieval
  - Reduced redundancy

Testing and Debugging
- Testing was performed after integrating all modules.
- Testing activities included:
  - Functional testing
  - API testing
  - User interface testing
  - Responsiveness testing
  - Database testing
  - Validation testing
- Debugging activities helped identify and resolve frontend and backend errors.
- The application was optimized for better performance and reliability.

Deployment and Documentation
- After testing, the project was prepared for deployment.
- Documentation activities included:
  - System architecture explanation
  - Module descriptions
  - Database design details
  - Workflow explanation
  - Testing reports
  - Screenshots and implementation results
- Proper documentation improved project organization and maintainability.

4.2 Detailed Description

Overview of the System
- The “BOOK MY TICKET ” is a full-stack web application.
- The application provides a centralized platform for:
  - Product management
  - Inventory tracking
  - Billing generation

- Customer handling
- Order processing
- Sales management

- The system reduces manual workload and improves operational efficiency.

Frontend Description

- The frontend was designed using modern web technologies.
- Main frontend responsibilities include:
    - Rendering product details and categories
    - Managing user interactions
    - Displaying shopping cart and billing interfaces
    - Handling navigation and routing
    - Sending API requests to the backend
    - Dynamically updating data on the interface
- State management and responsive design improved user experience.

Backend Description

- The backend handled all server-side operations and business logic.
- Major backend operations included:
    - Processing API requests
    - User authentication and authorization
    - Product and inventory management
    - Billing and order handling
    - Validation and exception management
    - Generating JSON responses
- RESTful APIs established communication between frontend and backend modules.

Database Description

- MySQL was used as the relational database management system.
- The database stored:
    - User details
    - Product information
    - Inventory records
    - Orders and billing details
    - Transaction history
- Important tables used in the system:
    - Users Table
    - Products Table
    - Orders Table
    - Inventory Table
- Database schemas were designed for efficient storage and retrieval.

User Authentication

- The system included secure authentication and authorization mechanisms.
- Authentication workflow:
    1. User registration using credentials
    2. Password encryption before storage
    3. Login credential verification
    4. Secure session/token generation

5. Authorization verification for protected routes

- These features improved application security and protected business data.

Product and Inventory Management

- This module handled:
    - Product addition and updates
    - Category management
    - Inventory tracking
    - Stock availability management
    - Product removal
- Inventory records were updated automatically whenever products were purchased or modified.

Billing and Order Management

- The application automated billing and order handling processes.
- Features included:
    - Automatic bill generation
    - Customer order management
    - Purchase history tracking
    - Sales and transaction recording
    - Billing management
- This module reduced manual errors and improved operational efficiency.

Responsive Design

- The application supported multiple devices and screen sizes.
- Responsive design features included:
    - Flexible layouts
    - Mobile-friendly navigation
    - Adaptive UI components
    - Improved accessibility and usability
- The responsive interface improved customer convenience and user experience.

System Workflow

- The application followed a client-server architecture.
- Workflow of the system:
    1. Users access the frontend interface.
    2. Requests are sent to the backend server.
    3. The backend processes requests and validations.
    4. The backend communicates with the database.
    5. The database stores or retrieves information.
    6. Responses are returned to the frontend.
    7. The frontend displays updated information dynamically.
- This workflow ensured smooth communication, better performance, and efficient data management.

PART I – BOOK MY TICKET SYSTEM

The “BOOK MY TICKET System” is a full-stack web application developed to provide a centralized and automated platform for managing ticket booking operations efficiently. The project was developed during the internship at JSpiders using modern web development technologies and software engineering concepts. The system was designed to simplify traditional ticket booking activities such as schedule management, seat availability tracking, ticket reservation, billing generation, customer handling, and booking management through a secure and user-friendly digital interface.

The project was developed using a three-tier architecture consisting of frontend, backend, and database layers. The frontend layer handles user interaction and interface rendering, the backend layer processes business logic and API communication, and the database layer stores and manages application data efficiently. This architecture ensures scalability, maintainability, and smooth communication between different components of the application.

4.1 Frontend Development

The frontend of the BOOK MY TICKET System was developed to provide an interactive, responsive, and user-friendly interface for both administrators and users. Modern frontend technologies such as HTML, CSS, Bootstrap, and JavaScript were used to design the user interface and improve user experience.

Features Implemented in Frontend

- Responsive homepage design
- Navigation bar and routing
- Ticket search interface
- Schedule display interface
- Seat selection and booking UI
- User registration and login forms
- Booking and checkout pages
- Admin dashboard interfaces
- Dynamic content rendering
- Responsive layouts for mobile and desktop devices

Technologies Used

- HTML5 for webpage structure
- CSS3 for styling and layout design
- Bootstrap for responsive UI components
- JavaScript for dynamic interactions and validations

Frontend Functionalities

- Dynamic schedule rendering
- Form validation
- User interaction handling
- API request handling
- Real-time interface updates
- Responsive navigation system

The frontend communicates with backend APIs to fetch and display booking data dynamically based on user actions.

4.2 Backend Development

The backend of the application acts as the core processing layer responsible for implementing business logic, managing requests, handling authentication, and communicating with the database. The backend was developed using Java, Spring Boot, and Hibernate/JPA.

Backend Features

• RESTful API development
• User authentication and authorization
• Schedule management APIs
• Booking and reservation processing
• Seat availability management
• Customer management
• Validation and exception handling
• Secure session management

Technologies Used

• Java for backend programming
• Spring Boot for application framework
• Hibernate/JPA for ORM mapping
• REST APIs for communication
• Maven for dependency management

Backend Operations

• Processing user booking requests
• Handling server-side business logic
• Validating input data
• Managing transactions
• Generating JSON responses
• Maintaining application security

The backend ensures smooth communication between frontend modules and the database.

4.3 Database Management

The database layer was designed to store and manage application data efficiently using MySQL. Database schemas and relationships were implemented carefully to maintain consistency and efficient retrieval of information.

Database Tables

• Users Table
• Schedules Table
• Routes Table
• Bookings Table
• Payments Table
• Transactions Table

Database Features

• Secure data storage
• Relational schema design
• CRUD operations
• Booking tracking
• Transaction management
• Data consistency maintenance

Database Functionalities

• Storing user credentials
• Managing travel schedules and routes
• Recording bookings and payments
• Updating seat availability automatically
• Maintaining customer booking history

Hibernate/JPA was used for object-relational mapping between Java classes and database tables.

4.4 User Authentication System

The application includes secure authentication and authorization mechanisms to protect sensitive booking data and control access to system functionalities.

Authentication Features

• User registration
• Secure login system
• Password encryption
• Role-based authorization
• Session/token management
• Protected routes

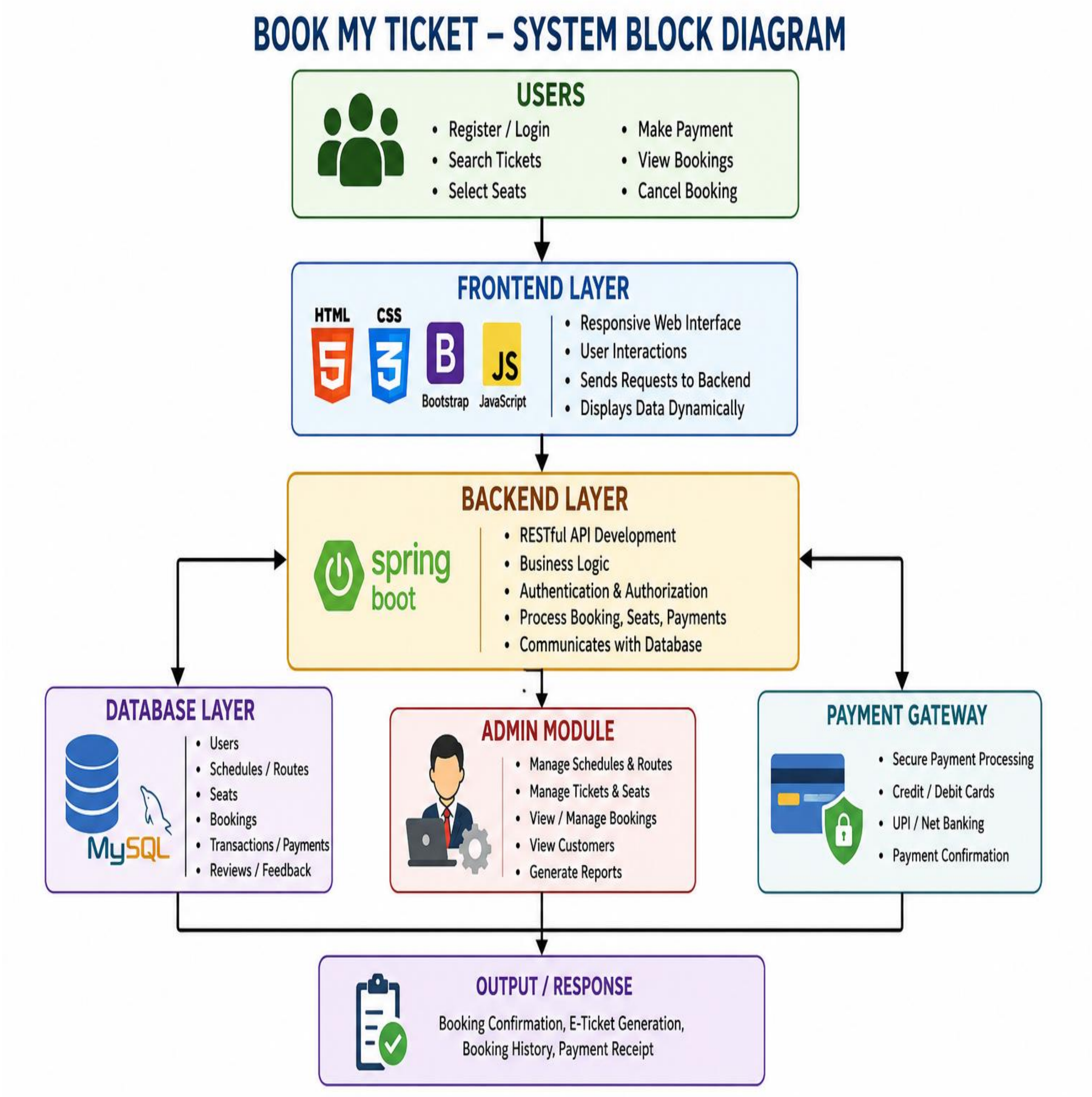
Authentication Workflow

1. User registers with credentials
2. Passwords are encrypted before storage
3. Login credentials are verified
4. Authentication session/token is generated
5. Authorized access is granted based on roles

The authentication system ensures secure access for administrators and users.

CHAPTER 5

BOOK MY TICKET System



5.1 Introduction

The “BOOK MY TICKET System” is a full-stack web application developed to automate and simplify ticket booking operations in modern transportation systems. Traditional booking methods often lead to delays, errors, and inefficiencies due to manual processing. To overcome these issues, this system provides a centralized digital platform for managing schedules, seat availability, bookings, payments, and customer records efficiently.

The application allows users to search routes, check availability, book tickets, and manage reservations through a simple and responsive interface. Administrators can manage schedules, bookings, and customer data effectively. The system reduces manual effort, improves accuracy, and enhances user experience.

It is built using a three-tier architecture:

- Frontend: HTML, CSS, Bootstrap, JavaScript
- Backend: Java, Spring Boot, Hibernate/JPA
- Database: MySQL

The system also includes secure authentication, role-based access, and RESTful API integration for smooth communication between components.

Overall, the project demonstrates a complete and efficient solution for digital ticket booking using modern web technologies.

5.2 Key Technologies Used

Frontend

- HTML5 – Page structure
- CSS3 – Styling and design
- Bootstrap – Responsive layout
- JavaScript – Interactivity and validation

Backend

- Java – Business logic
- Spring Boot – REST API development
- Hibernate/JPA – Database mapping

Database

- MySQL – Stores users, bookings, schedules, payments, and transactions

Additional Tools

- REST APIs – Frontend-backend communication
- JWT/Session – Authentication
- Maven – Project management
- Responsive Design – Mobile-friendly UI

5.3 Expected Outcomes

- User-friendly ticket booking system

- Admin dashboard for managing schedules and bookings
- Real-time seat availability updates
- Automated booking and payment processing
- Secure authentication and role-based access
- Responsive design for all devices
- Reduced manual work and improved efficiency
- Better user experience with fast booking process
- Practical implementation of full-stack development concepts

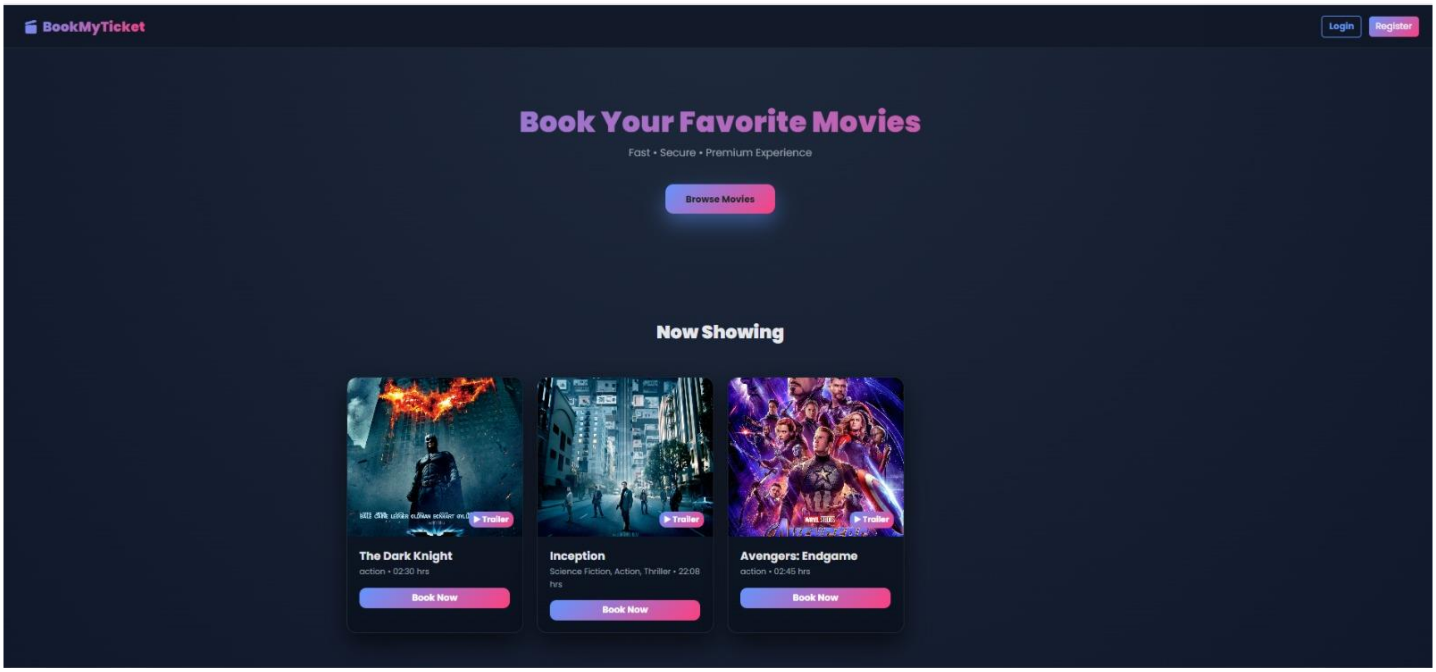


Fig 5.1 :DASHBOARD

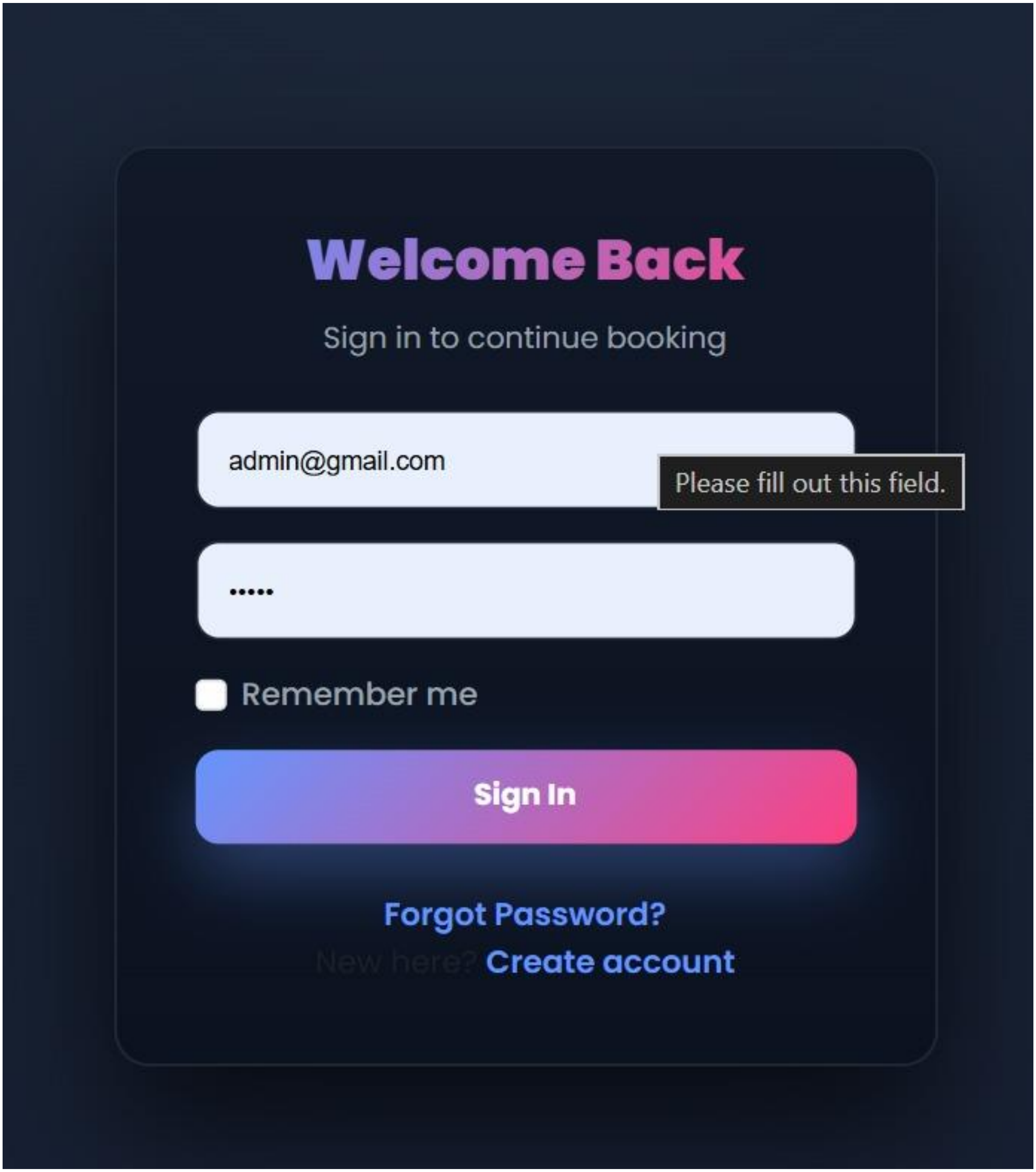
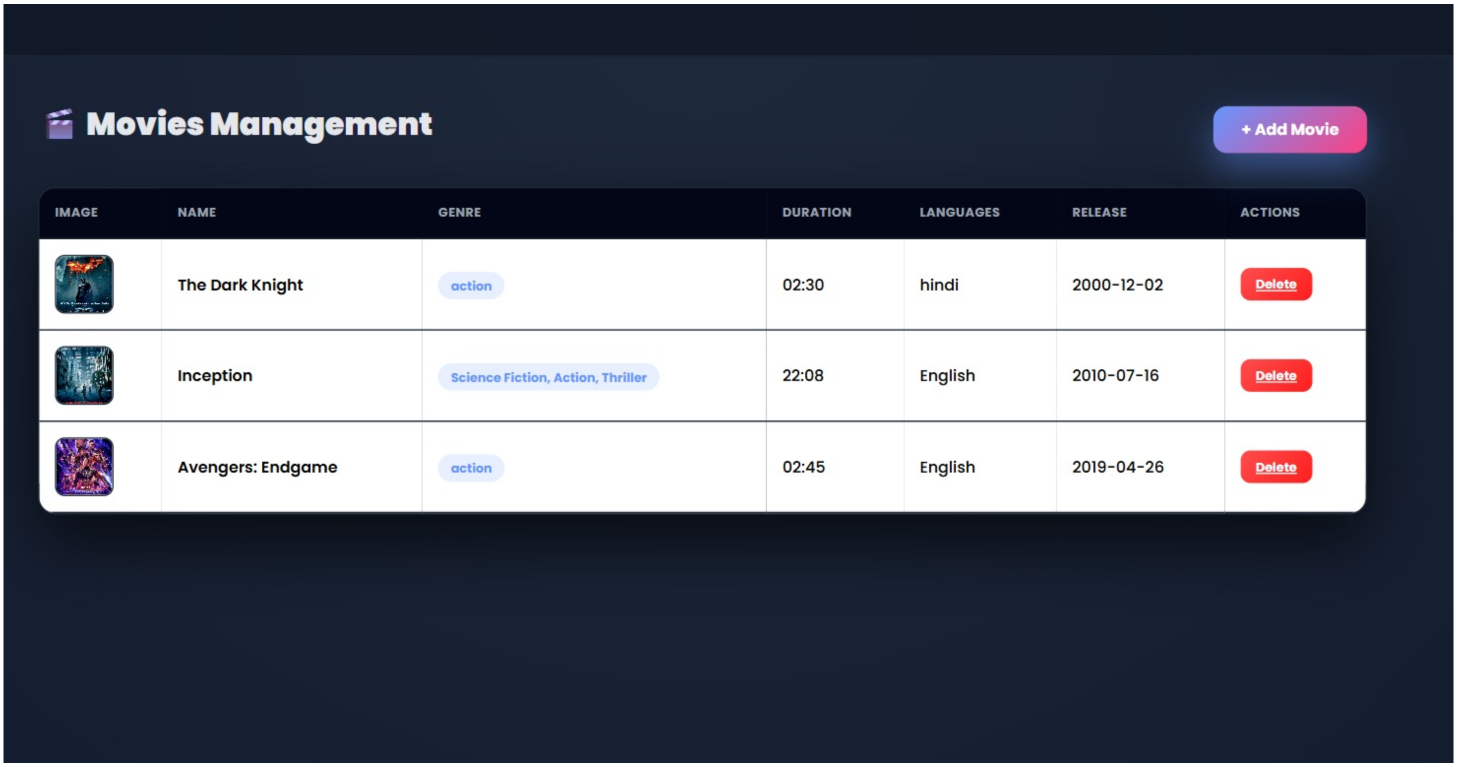


Fig 5.2 :ADMIN DASHBOARD


Movies Management
+ Add Movie
IMAGE
NAME
GENRE
DURATION
LANGUAGES
RELEASE
ACTIONS
The Dark Knight
action
02:30
hindi
2000-12-02
Delete
Inception
Science Fiction, Action, Thriller
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Avengers: Endgame
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02:45
English
2019-04-26
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Fig 5.3

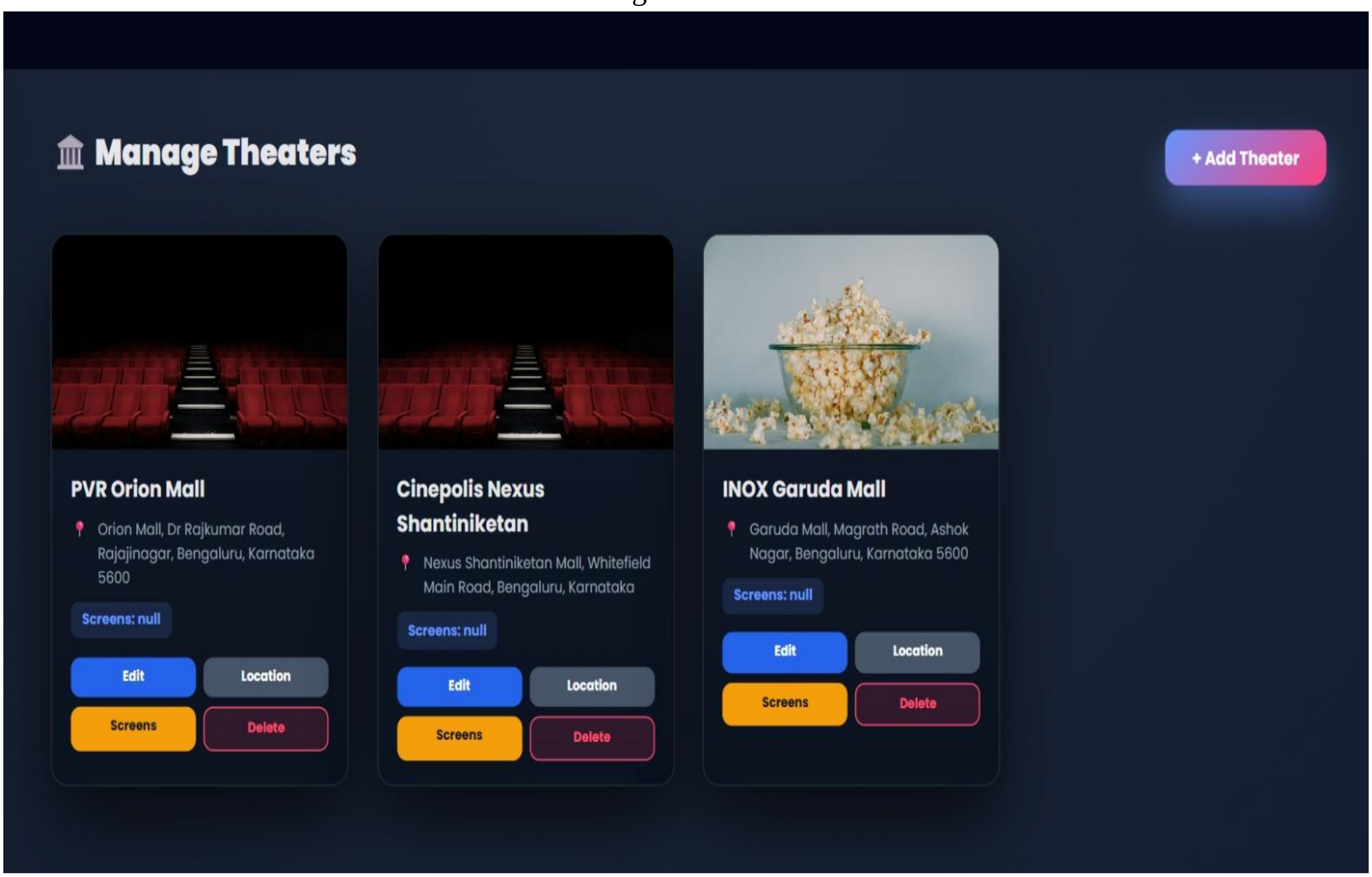
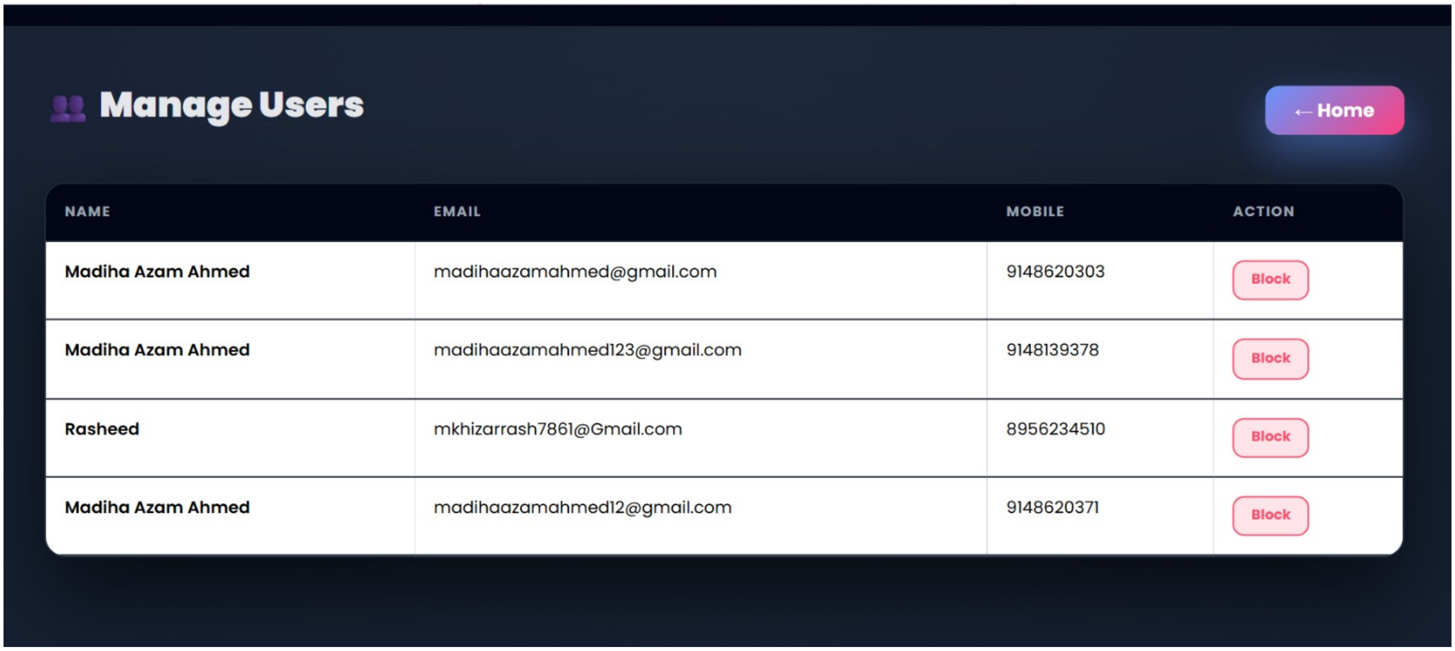

Manage Theaters
+ Add Theater
PVR Orion Mall
Orion Mall, Dr Rajkumar Road, Rajajinagar, Bengaluru, Karnataka 5600
Screens: null
Edit
Location
Screens
Delete
Cinepolis Nexus Shantiniketan
Nexus Shantiniketan Mall, Whitefield Main Road, Bengaluru, Karnataka
Screens: null
Edit
Location
Screens
Delete
INOX Garuda Mall
Garuda Mall, Magrath Road, Ashok Nagar, Bengaluru, Karnataka 5600
Screens: null
Edit
Location
Screens
Delete

Fig 5.4

Fig 5.5


Manage Users
← Home
NAME
EMAIL
MOBILE
ACTION
Madiha Azam Ahmed
madihaazamahmed@gmail.com
9148620303
Block
Madiha Azam Ahmed
madihaazamahmed123@gmail.com
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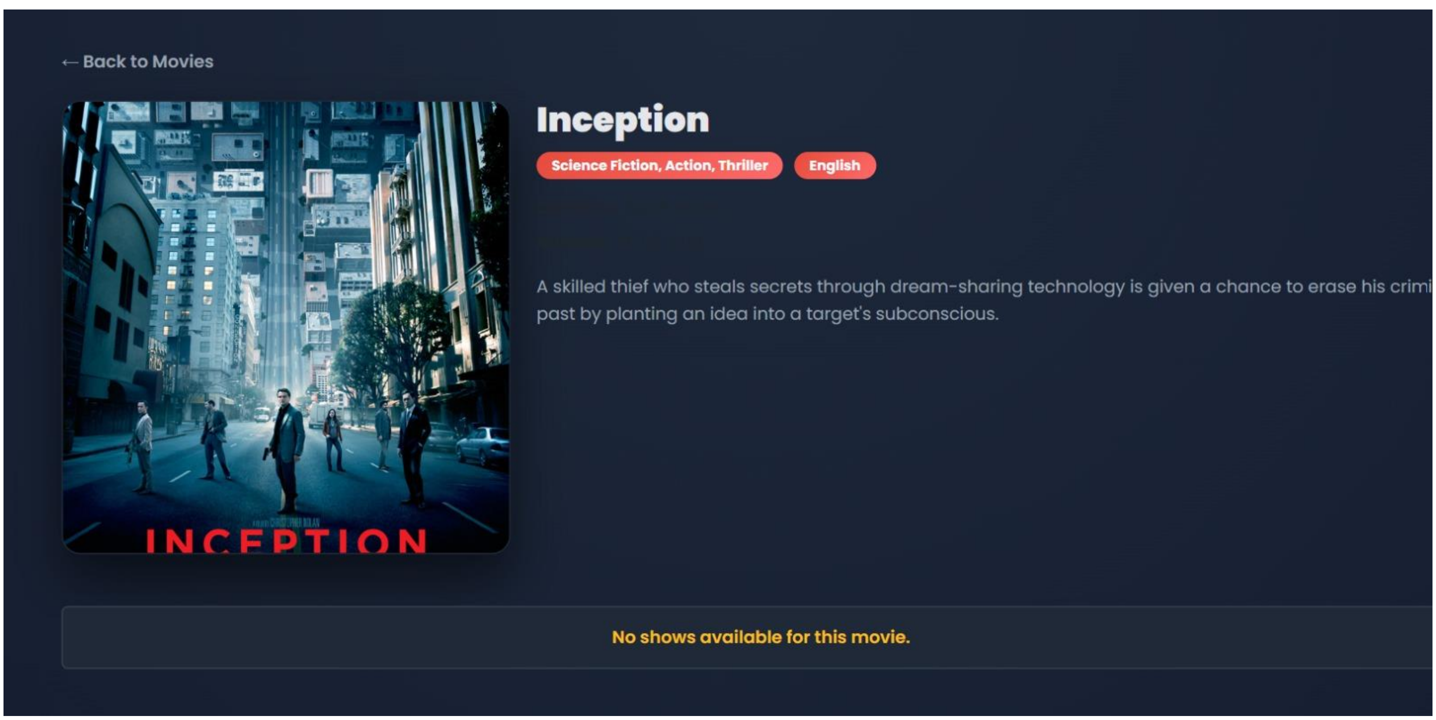
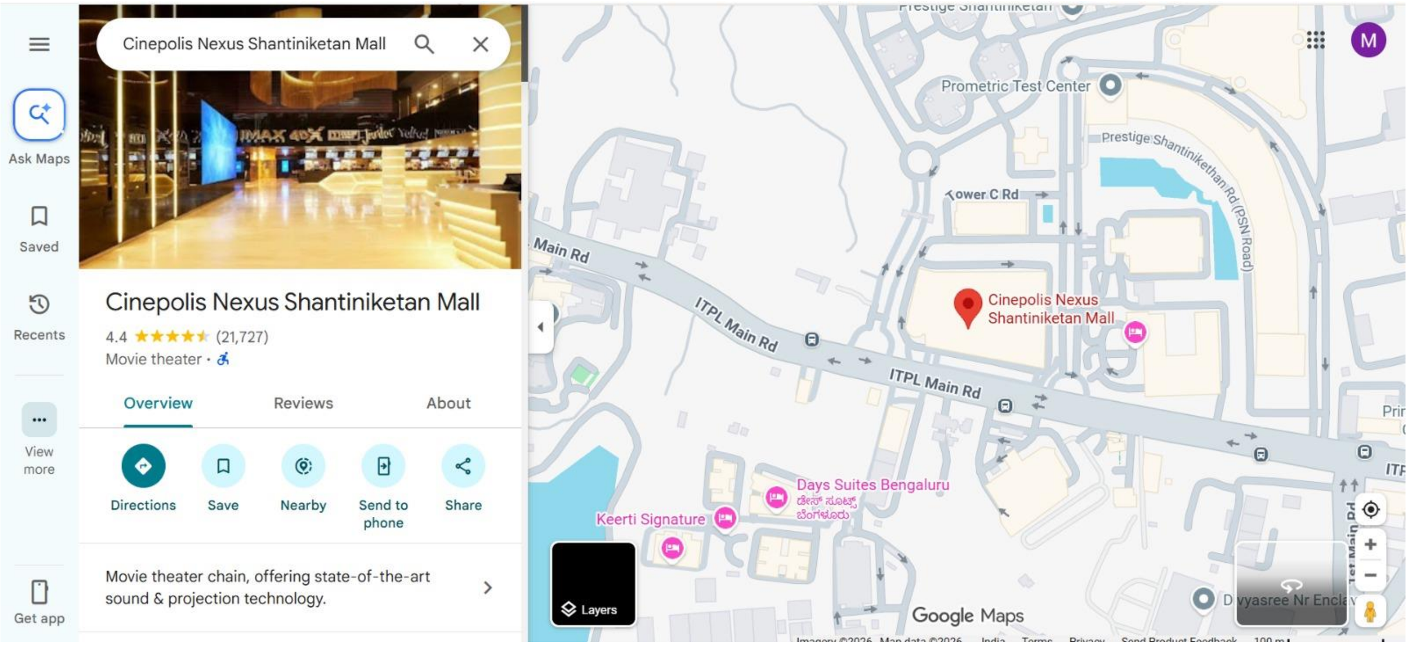

← Back to Movies
Inception
Science Fiction, Action, Thriller
English
A skilled thief who steals secrets through dream-sharing technology is given a chance to erase his crimi
past by planting an idea into a target's subconscious.
INCEPTION
No shows available for this movie.

FIG 5.6



5.1 PROPOSED SYSTEM

The proposed "BOOK MY TICKET" system is a web-based online ticket booking application designed to automate and simplify the ticket reservation process for users and administrators. The system provides a centralized platform where customers can search available tickets, check schedules, select seats, make online payments, and receive instant booking confirmations efficiently. The application offers separate modules for users and administrators to ensure secure and organized management of booking operations. Users can register and log in securely, browse available tickets, search based on date, destination, movie/event name, or category, and book tickets through a simple and user-friendly interface. The system also allows users to view booking history, download tickets, and cancel reservations when required.

The Admin Module is designed for administrators and shop managers to monitor and manage business operations effectively. Administrators can add new products, update product information, manage categories, remove unavailable products, track inventory levels, and monitor customer orders in real time. The admin dashboard also provides access to customer records, billing details, sales information, and transaction history. By centralizing these functionalities into a single system, the application reduces manual effort and improves operational accuracy and efficiency.

For administrators, the system provides functionalities to manage schedules, ticket availability, seat allocation, customer records, booking details, and payment information. Administrators can add, update, or remove schedules and monitor all booking activities through a centralized dashboard.

The proposed system includes real-time seat availability and automated ticket generation to avoid double bookings and manual errors. Secure authentication and online payment integration ensure safe transactions and protection of user data. The system also generates booking reports and maintains transaction records for efficient management and analysis. The application is designed with a responsive interface to support accessibility across desktops, laptops, tablets, and smartphones. Modern web technologies and database management techniques are used to ensure scalability, reliability, and fast performance.

Overall, the proposed BOOK MY TICKET system provides a secure, efficient, and user-friendly digital solution for managing ticket reservations, improving customer convenience, reducing manual workload, and enhancing operational efficiency.maintain data consistency, faster retrieval, and organized storage. Hibernate/JPA is used for object-relational mapping and smooth interaction between the backend and database layers.

EXISTING SYSTEM

In the existing ticket booking system, most operations are handled manually or through semi-digital methods such as offline counters, handwritten records, and basic reservation tools. Customers need to visit booking centers or rely on limited online platforms to check availability and book tickets.

The existing system faces several limitations such as delayed booking confirmation, lack of real-time seat availability updates, manual errors in ticket generation, and difficulty in maintaining accurate passenger and transaction records. It also requires more human effort for managing schedules, cancellations, and payments.

Since data is not fully centralized, it becomes difficult for administrators to track bookings, manage schedules efficiently, and generate reports. The system also lacks flexibility, scalability, and user-friendly interfaces for modern users.
Overall, the existing system is inefficient, time-consuming, and not suitable for handling large-scale ticket booking operations in today's digital environment.

Existing System

- The existing system depends heavily on manual work, which increases the chances of human errors in booking and record maintenance.
- There is no proper real-time synchronization of seat availability, leading to issues like overbooking or booking conflicts.
- Customers often face long waiting times at ticket counters during peak hours.
- The system lacks proper data security, making customer and transaction information vulnerable.
- Report generation is slow and requires manual calculation, which reduces efficiency.
- There is no centralized database, so data is scattered and difficult to manage.
- Limited accessibility, as users must physically visit booking centers or depend on restricted platforms.
- The system is not scalable to handle a large number of users and booking requests efficiently.

7.1 Functional Modules

The "BOOK MY TICKET System" is divided into several functional modules to ensure smooth and organized management of ticket booking operations. Each module performs specific tasks within the application. These modules work together to provide a centralized, efficient, and user-friendly ticket booking platform. The major functional modules include User Authentication Module, Schedule Management Module, Booking and Seat Management Module, Customer Management Module, Payment and Billing Module, Admin Module, and Reporting Module.

1. User Authentication Module

The User Authentication Module provides secure access to the system for users and administrators. It ensures that only authorized users can access the application. Users can register and log in using secure credentials, and the system verifies identity before granting access.

Main Functions:

- User registration and login
- Password encryption and verification
- Session/token management
- Role-based access control
- Secure access to system features

2. Schedule Management Module

The Schedule Management Module allows administrators to manage travel schedules, routes, and timings efficiently. It ensures updated and accurate availability of travel options.

Main Functions:

- Add and update schedules
- Manage routes and timings
- Maintain seat availability
- Delete inactive schedules
- View schedule details

3. Booking and Seat Management Module

This module handles ticket reservation and seat allocation. Users can book tickets while the system updates seat availability in real time.

Main Functions:

- Ticket booking and cancellation
- Seat selection and allocation
- Real-time availability updates
- Booking confirmation
- Booking history tracking

4. Customer Management Module

The Customer Management Module manages user details and booking history. It helps maintain proper customer records and improves service efficiency.

Main Functions:

• Customer registration and profile management
• View booking history
• Maintain user details
• Manage customer activities
• Improve user experience

5. Payment and Billing Module

This module automates ticket payment and billing operations. It ensures accurate transaction processing and secure payment handling.

Main Functions:

• Automatic ticket generation
• Payment processing
• Transaction recording
• Billing management
• Payment history tracking

6. Admin Module

The Admin Module provides complete control over the system. Administrators manage schedules, bookings, users, and system activities through a centralized dashboard.

Main Functions:

• Manage users and schedules
• Monitor bookings and transactions
• View reports and analytics
• Manage system settings
• Control overall operations

7. Reporting Module

The Reporting Module helps in analyzing booking performance and system usage. It supports better decision-making through structured reports.

Main Functions:

• Generate booking reports
• View transaction history
• Analyze system performance
• Track revenue and usage
• Maintain records

8. Responsive User Interface Module

The Responsive UI Module ensures the application works smoothly across all devices like mobiles, tablets, and desktops.

Main Functions:

• Responsive web design
• Mobile-friendly interface

- Smooth navigation
- Dynamic UI updates
- Better accessibility and usability

CHAPTER 6

ANALYSIS

6.1 Introduction

The analysis chapter presents a systematic examination of the technologies, systems, and workflows studied and implemented during the internship. This analysis aims to contextualize the learning outcomes within the broader landscape of software engineering and data science practice, evaluate the relative strengths and applicability of the technologies covered, and identify the key insights gained through hands-on implementation.

The analysis chapter presents a systematic study of the technologies, system architecture, and workflow used in the development of the "BOOK MY TICKET System." It explains how different components of the application interact to form a complete and efficient ticket booking platform. This chapter also highlights the practical knowledge gained during the internship at JSpiders through hands-on implementation of full-stack web development concepts.

The BOOK MY TICKET System was developed to automate ticket booking operations such as schedule management, seat availability tracking, booking processing, customer management, and payment handling. The system integrates frontend, backend, and database technologies to provide a centralized and efficient digital solution.

The analysis focuses on understanding how modern web technologies work together in real-world applications. In full-stack development, applications are built using multiple layers rather than a single technology. This project follows the same approach by combining frontend interfaces, backend services, and database systems.

Frontend development is used to design interactive and responsive user interfaces using HTML, CSS, Bootstrap, and JavaScript. Backend development handles business logic, authentication, booking processing, and API communication using Java, Spring Boot, and Hibernate/JPA. The database layer manages and stores schedules, bookings, payments, and user data using MySQL.

6.2 System Overview

The "BOOK MY TICKET System" is a full-stack web-based application designed to automate and manage ticket booking operations in an efficient and user-friendly manner. The system provides a centralized platform for handling schedules, seat availability, ticket reservations, payments, and customer information through a single digital interface.

The main objective of the system is to eliminate manual booking processes and provide a fast, accurate, and secure online ticket reservation solution. It enables users to search for available routes, view schedules, select seats, book tickets, and make payments easily. Administrators can manage schedules, monitor bookings, and maintain system data efficiently.

The application is developed using a three-tier architecture consisting of:

- **Frontend:** Built using HTML, CSS, Bootstrap, and JavaScript for creating responsive and interactive user interfaces.
- **Backend:** Developed using Java, Spring Boot, and Hibernate/JPA to handle business logic, authentication, and API communication.
- **Database:** MySQL is used for storing user data, schedules, bookings, payments, and transaction records securely.

The system also includes secure authentication and role-based access control to ensure that only authorized users can access specific functionalities. RESTful APIs are used for smooth communication between frontend and backend modules.

6.3 Modules of the System

The “BOOK MY TICKET System” is divided into several functional modules that work together to provide an efficient and organized ticket booking platform. Each module is responsible for specific operations such as user management, schedule handling, booking, payment processing, and system administration.

1. User Authentication Module

This module manages user registration, login, password verification, and authorization. It ensures secure access to the system by validating user credentials and protecting sensitive information.

2. Schedule Management Module

The Schedule Management Module allows administrators to add, update, and manage travel schedules, routes, timings, and seat availability. It ensures that users always get updated schedule information.

3. Booking and Seat Management Module

This module handles ticket booking, seat selection, cancellation, and real-time seat availability updates. It ensures smooth and accurate reservation processing.

4. Customer Management Module

The Customer Management Module stores and manages user details, booking history, and account information. It helps in maintaining proper customer records and improving service efficiency.

5. Payment and Billing Module

This module manages ticket payment processing, bill generation, and transaction records. It ensures secure and automated payment handling for all bookings.

6. Admin Dashboard Module

The Admin Dashboard provides complete control over the system. Administrators can manage users, bookings,payments, and view system activities through a centralized interface.

7. Reporting Module

This module generates booking reports, transaction summaries, and performance analytics. It helps administrators analyze system usage and improve decision-making.

6.4 Technologies Used

The BOOK MY TICKET System is developed using modern full-stack web development technologies to ensure performance, scalability, and security.

Frontend Technologies

HTML5

Used to structure web pages such as booking forms, schedule pages, and dashboards.

CSS3

Used for styling the application and improving layout, design, and responsiveness.

JavaScript

Used for interactivity, form validation, and dynamic updates.

Bootstrap

Used to create responsive and mobile-friendly user interfaces.

Backend Technologies

Java

Used to implement business logic and core application functionality.

Spring Boot

Used for building RESTful APIs and backend services.

Hibernate/JPA

Used for database mapping and data persistence.

REST APIs

Used for communication between frontend and backend systems.

Database Technology

MySQL

Used for storing user data, schedules, bookings, payments, and transaction records securely.

SQL

Used for performing CRUD operations and managing database queries.

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6.4 WORKING OF THE SYSTEM – BOOK MY TICKET SYSTEM

The "BOOK MY TICKET System" works on a client-server architecture where the frontend, backend, and database work together to provide a smooth and automated ticket booking experience.

1. User Access
The user first accesses the application through a web browser. The homepage displays available options such as viewing schedules, searching routes, logging in, and booking tickets.

2. User Authentication
Users register by providing their details and then log in using secure credentials. The system verifies the username and password through the backend. If the credentials are valid, the user is granted access based on their role (user or admin).

3. Schedule and Seat Availability
After login, users can view available travel schedules and routes. The system fetches real-time data from the database and displays seat availability, timings, and fare details.

4. Ticket Booking Process
The user selects a route, chooses available seats, and proceeds to book tickets. The booking request is sent from the frontend to the backend using REST APIs.

5. Backend Processing
The backend (Spring Boot) processes the request, validates data, checks seat availability, and performs business logic. If seats are available, the booking is confirmed.

6. Database Operations
All booking details, user data, schedule information, and payment records are stored in the MySQL database. The database is updated automatically after every booking or cancellation.

7. Payment and Ticket Generation
After successful booking, the system processes payment details and generates a ticket confirmation. The ticket information is stored and made available to the user.

8. Admin Operations
Administrators log in to a separate dashboard where they can manage schedules, users, bookings, and view reports. They also monitor system performance and update data when required.

9. Response Display
The backend sends the processed response back to the frontend, and the interface is updated dynamically to show booking confirmation, ticket details, or error messages.

Overall Working Flow:
User → Frontend → Backend → Database → Backend → Frontend
This workflow ensures fast, secure, and efficient ticket booking with real-time updates and minimal manual effort.

CONCLUSION

The “BOOK MY TICKET System” is a successfully developed full-stack web application designed to automate and simplify ticket booking operations. The system effectively replaces traditional manual booking methods with a centralized digital platform that improves speed, accuracy, and efficiency in managing tickets, schedules, users, and payments.The application provides a user-friendly interface where users can easily search routes, check seat availability, book tickets, and make secure payments. At the same time, administrators can efficiently manage schedules, bookings, and system data through a centralized dashboardBy using modern technologies such as Java, Spring Boot, Hibernate/JPA, MySQL, HTML, CSS, Bootstrap, and JavaScript, the system ensures scalability, security, and smooth performance. The implementation of RESTful APIs and role-based authentication further enhances system reliability and data protection.Overall, the project demonstrates the practical application of full-stack web development concepts learned during the internship. It provides a strong foundation for understanding real-world software development and helps improve technical, analytical, and problem-solving skills.

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